

SoundBar_SDK_On_21569

Flash Writer Guide

Version 1.0 April 2020



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Regulatory Compliance

The EVAL_Melody-8 is designed to be used solely in a laboratory environment. The board is not intended for use as a consumer end product or as a portion of a consumer end product. The board is an open system design which does not include a shielded enclosure and therefore may cause interference to other electrical devices in close proximity. This board should not be used in or near any medical equipment or RF devices.

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Preface

Thank you for purchasing the SoundBar SDK on EVAL-Melody-8 from Analog Devices, Inc. With a rich, powerful instruction set, floating point precision and high speed execution, it makes SHARC processors a popular choice for designers of Signal processing systems.

The evaluation board is designed to be used in conjunction with the -Cross Core® Embedded Studio (CCES) development tools environment and reference platform with an ADSP-2156x SHARC processor. The CCES development environment aids advanced application code development and debug, such as:

- Create, compile, assemble, and link application programs written in C++, C, and assembly
- Load, run, step, halt, and set breakpoints in application programs
- Read and write data and program memory
- Read and write core and peripheral registers

Access to the processor from a personal computer (PC) is achieved through an external JTAG emulator.

Analog Devices JTAG emulators offer faster communication between the Host PC and target hardware. Analog Devices carries a wide range of in-circuit emulation products. To learn more about Analog Devices emulators and processor development tools, go to <http://www.analog.com/dsp/tools/>.

Intended Audience

This manual is intended for use by a software team and customers, both externally and internally to Analog Devices, to develop and port codes on EVAL-Melody-8.

The primary audience for this manual is a programmer who is familiar with Analog Devices processors. This manual assumes that the audience has a working knowledge of the appropriate processor architecture, instruction set, and C/C++ programming languages.

Programmers who are unfamiliar with Analog Devices processors can use this manual, but should supplement it with other texts (such as the *ADSP-2156x SHARC Processor Hardware Reference* and the *ADSP-2156x SHARC Processor Programming Reference*) that describe your target architecture.

Programmers who are unfamiliar with Cross Core Embedded Studio should refer to the CCES online help.

What's New in This Manual

This is the first version of the SoundBar_SDK_On_21569_Flash_Write_Guide.

Technical Support

You can reach Analog Devices processors and DSP technical support in the following ways:

- Submit your questions to technical support directly at:
www.analog.com/support
- E-mail your questions about processors and processor applications to:
processor.support@analog.com or
processor.china@analog.com (Greater China support)
- Contact your Analog Devices sales office or authorized distributor. Locate one at:
www.analog.com/adi-sales
- Send questions by mail to:
Processors and DSP Technical Support
Analog Devices, Inc.
One Technology Way
P.O. Box 9106
Norwood, MA 02062
USA

Supported Processors

This evaluation system supports the Analog Devices ADSP-2156x SHARC processors.

Product Information

Product information can be obtained from the Analog Devices Web site and the CCES online help system.

Analog Devices Web Site

The Analog Devices Web site, www.analog.com, provides information about a broad range of products—analogue integrated circuits, amplifiers, converters, and digital signal processors.

To access a complete technical library for each processor family, go to www.analog.com/processors/technical_library. The manuals selection opens a list of current manuals related to the product as well as a link to the previous revisions of the manuals. When locating your manual title, note a possible errata check mark next to the title that leads to the current correction report against the manual.

Also note, MyAnalog.com is a free feature of the Analog Devices Web site that allows customization of a Web page to display only the latest information about products you are interested in. You can choose to receive weekly e-mail notifications containing updates to the Web pages that meet your interests, including documentation errata against all manuals. MyAnalog.com provides access to books, application notes, data sheets, code examples, and more.

Visit MyAnalog.com to sign up (found on the Analog Devices home page). If you are a registered user, just log on. Your user name is your e-mail address.

EngineerZone

EngineerZone is a technical support forum from Analog Devices. It allows you direct access to Analog Devices technical support engineers. You can search FAQs and technical information to get quick answers to your embedded processing and DSP design questions.

Use EngineerZone to connect with other DSP developers who face similar design challenges. You can also use this open forum to share knowledge and collaborate with the Analog Devices support team and your peers. Visit ez.analog.com to sign up.

1. Purpose

The objective of this document is to explain the flashing procedure of the application using the ADSP21569 with the serial flash W25Q128FV. The flashing on the hardware requires the following two steps.

- Creating the LDR file for the Code application software with the initialization code.
- Flash the above created LDR image to the serial flash.

For flashing the hardware, first CORE clocks and the DDR memory control registers need to be programmed. This will be done using the “21569_preload” & “21569_init” project. Based on the customer hardware DDR memory and the crystal, this code (“21569_preload” & “21569_init”) needs to be modified. The following steps explain the procedure to flash the serial flash to the ADSP2156x processor.

1.1 Development Environment

The following tools and accessories are required.

- EVAL-Melody8 board.
- CCES Version 2.9.1 build w1909051
- ICE-1000/2000 Emulator of Analog Devices

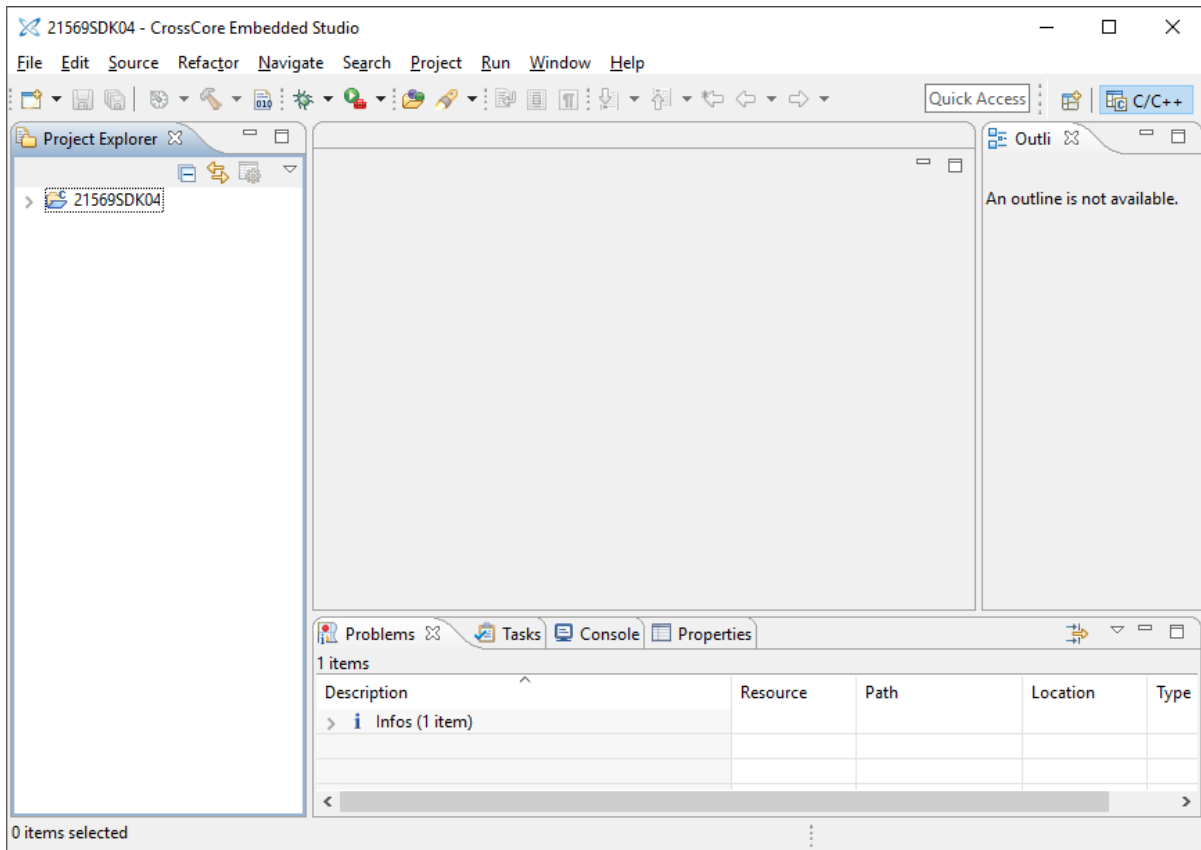
1.2 Loader Creation

The flashing on the hardware requires the following two steps.

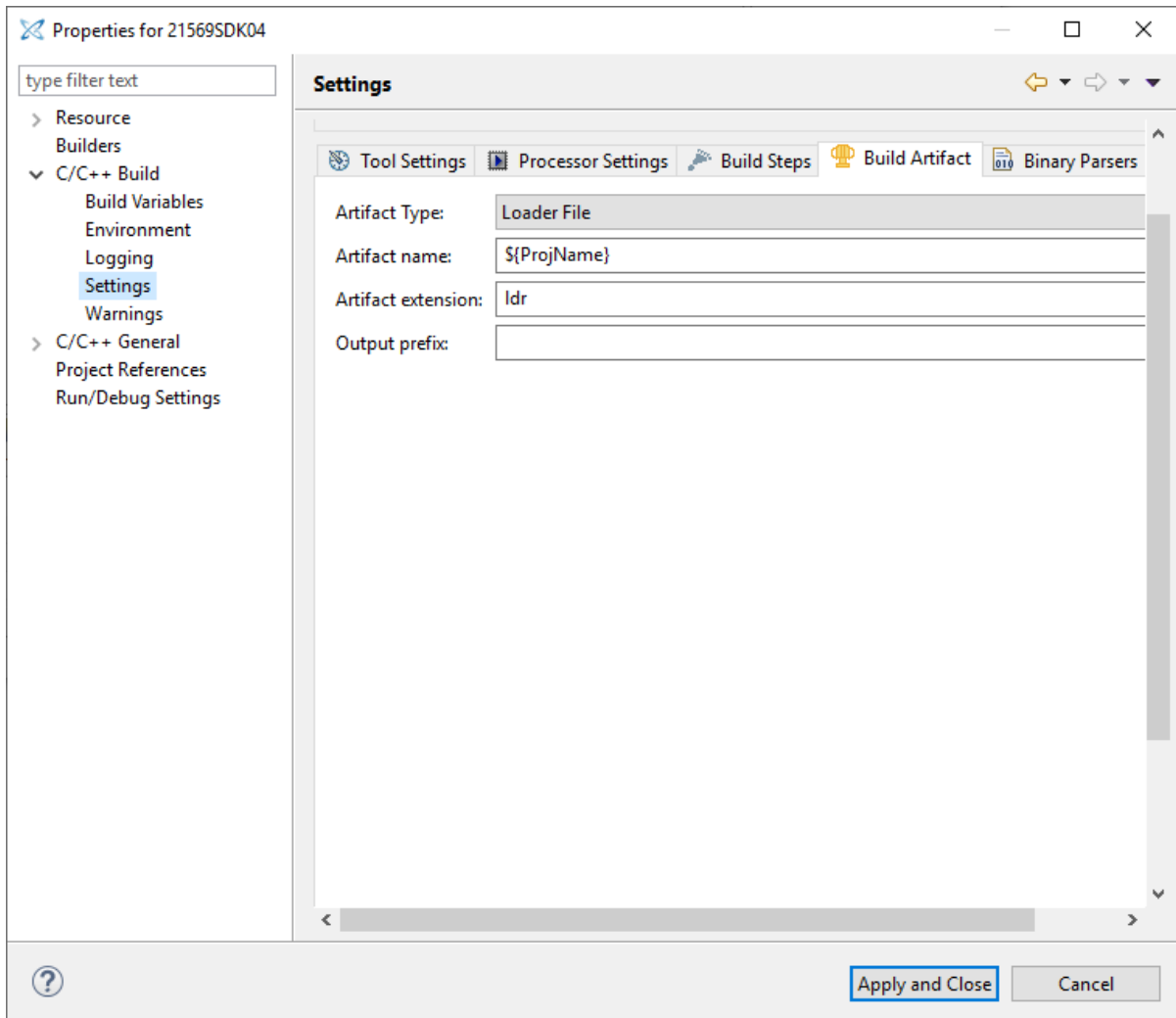
- Creating the LDR file for the Code application software with the initialization code.
- Flash the above created LDR image to the serial flash.

For flashing the hardware, first CORE clocks and the DDR memory control registers need to be programmed. This will be done using the “21569_preload” & “21569_init” project. Based on the customer hardware DDR memory and the crystal, this code (“21569_preload.dxe” & “21569_init”) needs to be modified. The following steps explain the procedure to flash the serial flash to the ADSP2156x processor

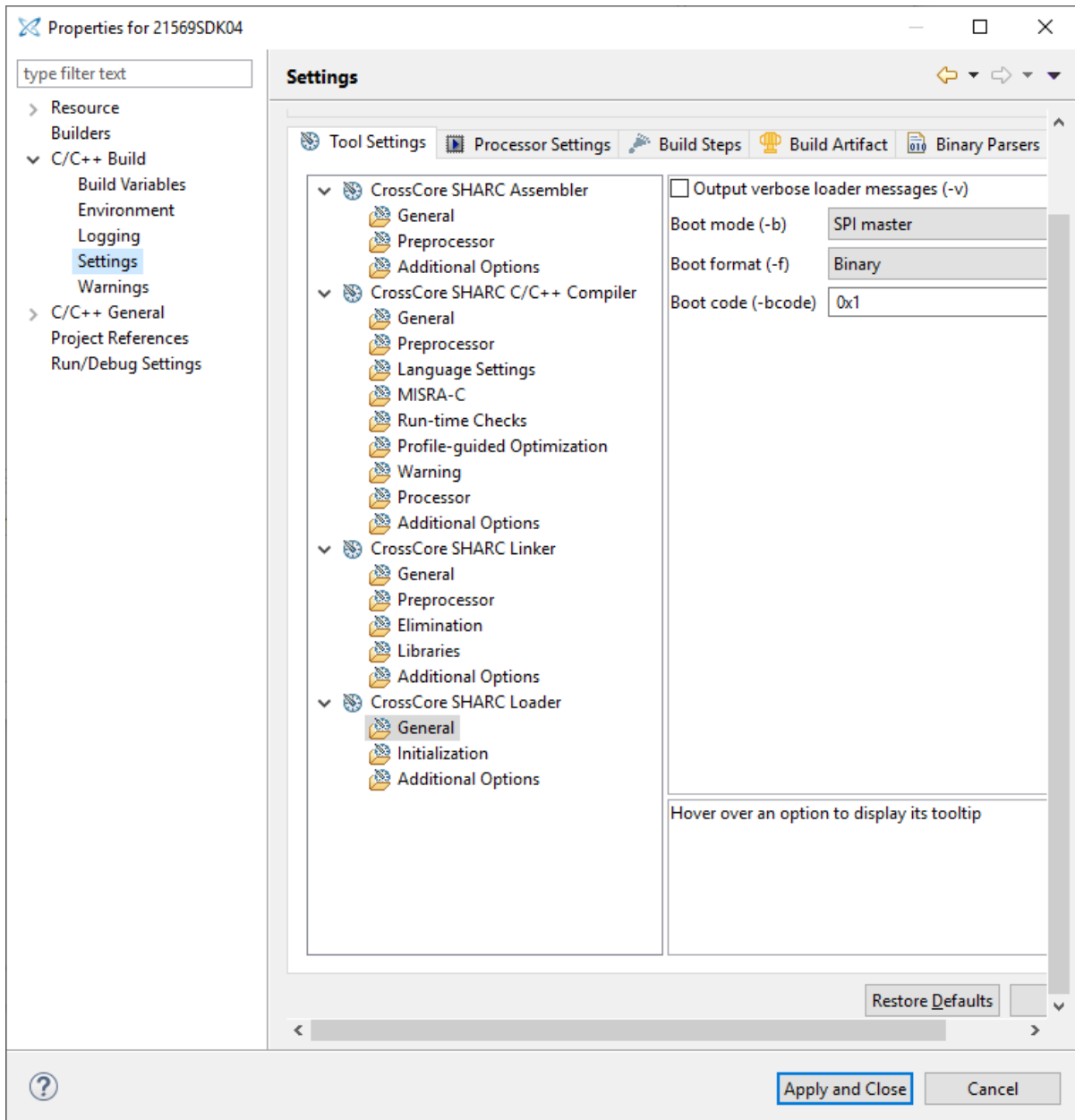
1. Import “21569SDK04” project to the CCES 2.9.1 workspace.



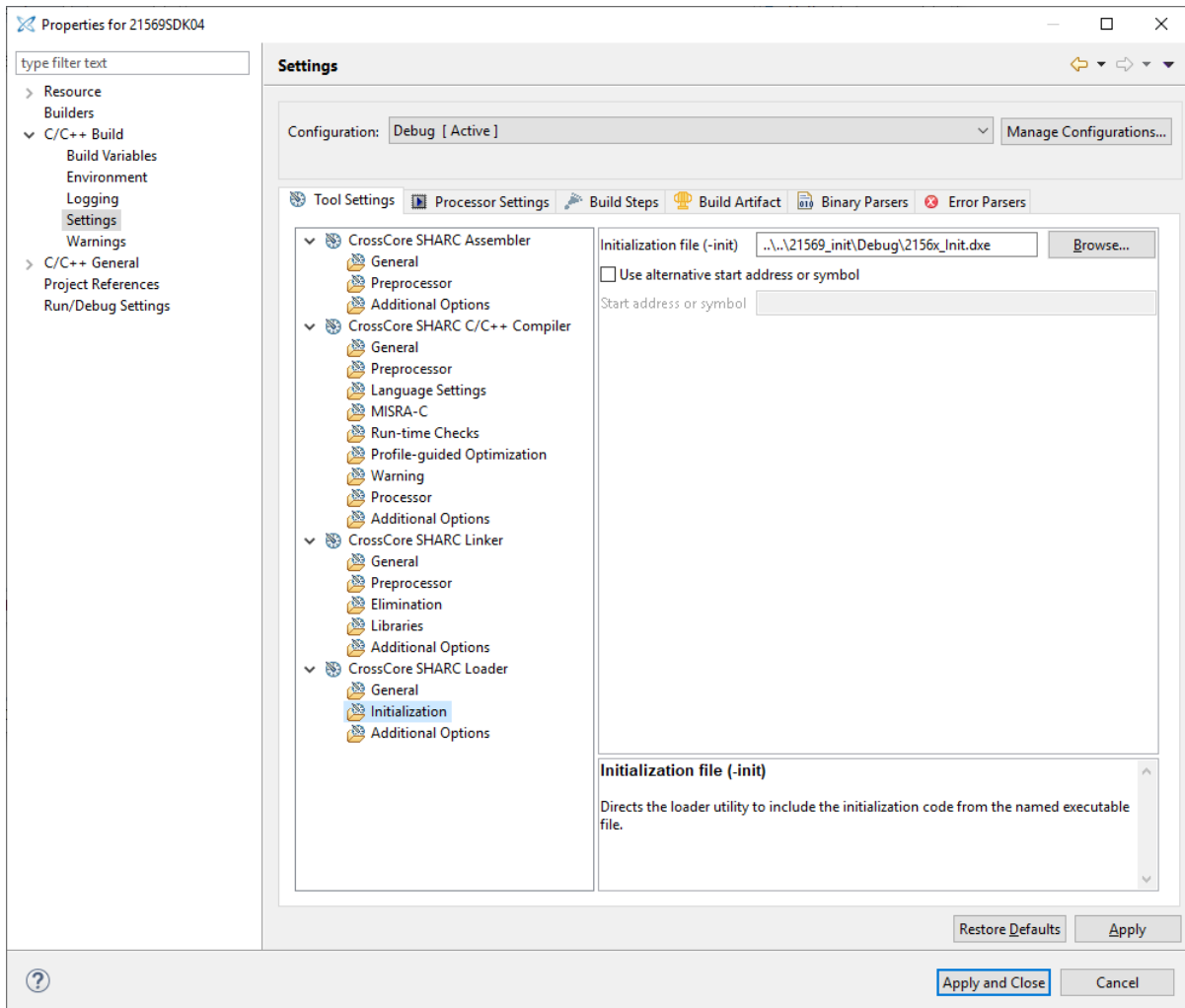
3. Go to Properties then C\C++ Build → Settings → Build Artifact displays as shown below,



4. In Artifact Type switch to the “Loader File”, then go back “Tool Settings”.
5. In CrossCore SHARC Loader → General, switch to “Binary” in Boot format and “0x1” in Boot Code.



6. In CrossCore SHARC Loader → Initialization, “..\2156x_Init.dxe”.



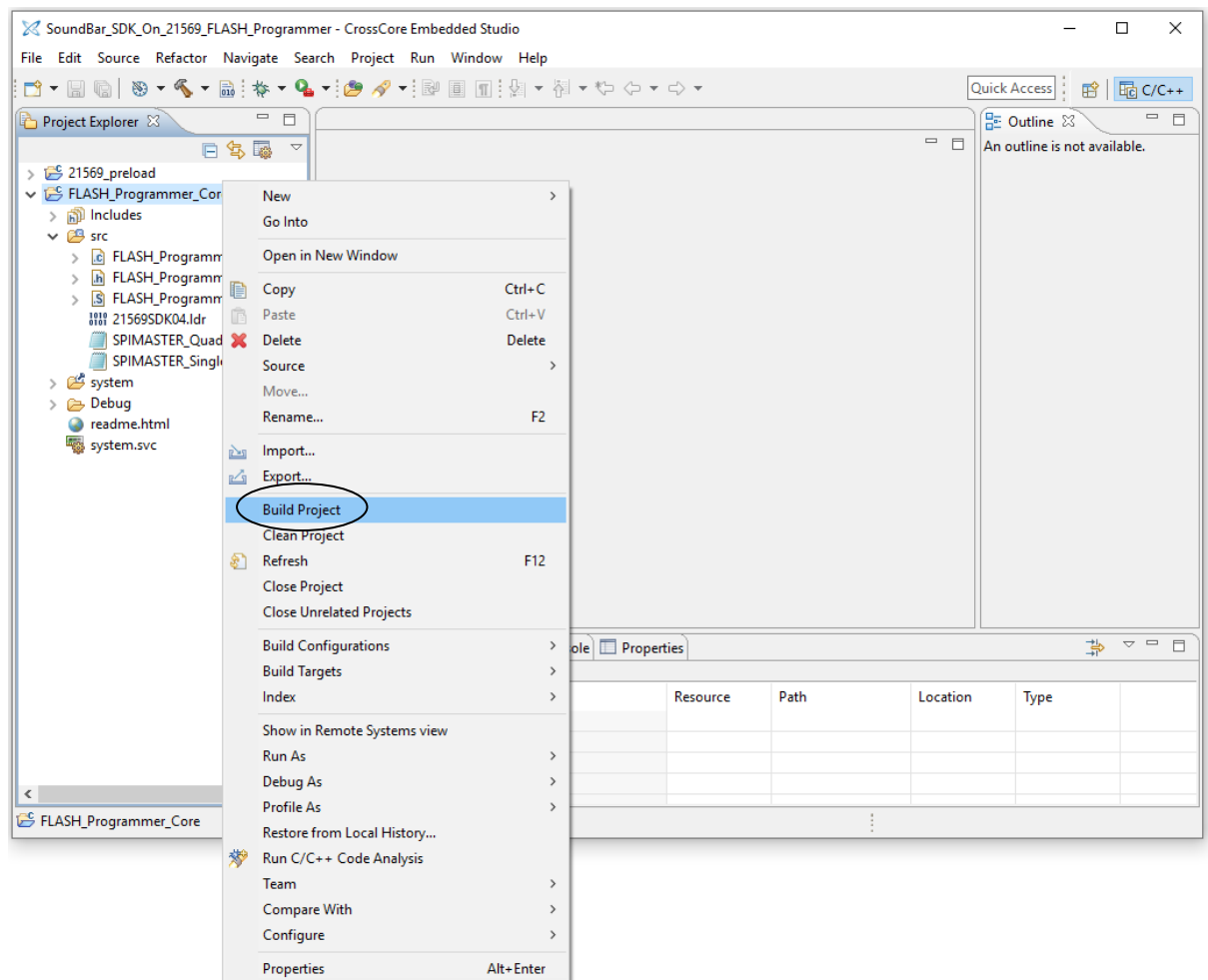
7. Press “Apply and Close”.
8. After building the “21569SDK04” project, the LDR file will be generated in its debug folder.

1.3 Flashing Procedure

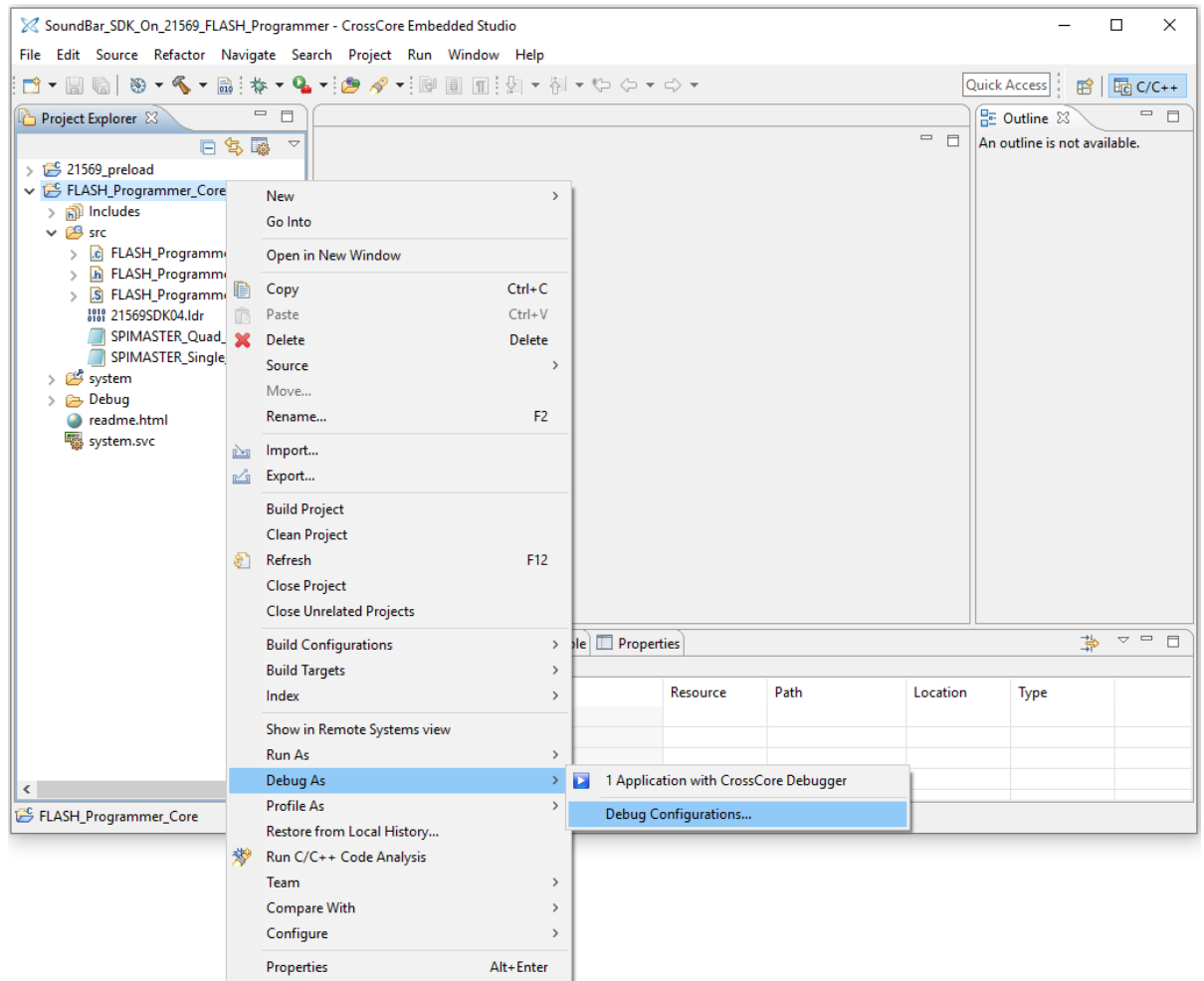
Step 1: Copy the Loader file from the Flashables folder to FLASH_Programmer_Core project “src” folder

Name	Date modified	Type	Size
21569SDK04.ldr	22-04-2020 14:35	LDR File	6,479 KB
FLASH_Programmer.asm	13-11-2019 18:54	Assembler Source	1 KB
FLASH_Programmer_Core1.c	19-11-2019 17:40	C Source	14 KB
FLASH_Programmer_Core1.h	11-05-2019 14:27	C/C++ Header	4 KB
SPIMASTER_Quad_MULTI_CORE.dat	30-10-2015 13:40	DAT File	794 KB
SPIMASTER_Single_ARM.dat	30-10-2015 13:39	DAT File	316 KB

Step 2: Import “FLASH_Programmer_Core” project to the CCES workspace (CCESv2.9.1 build w1909051) and build the project.

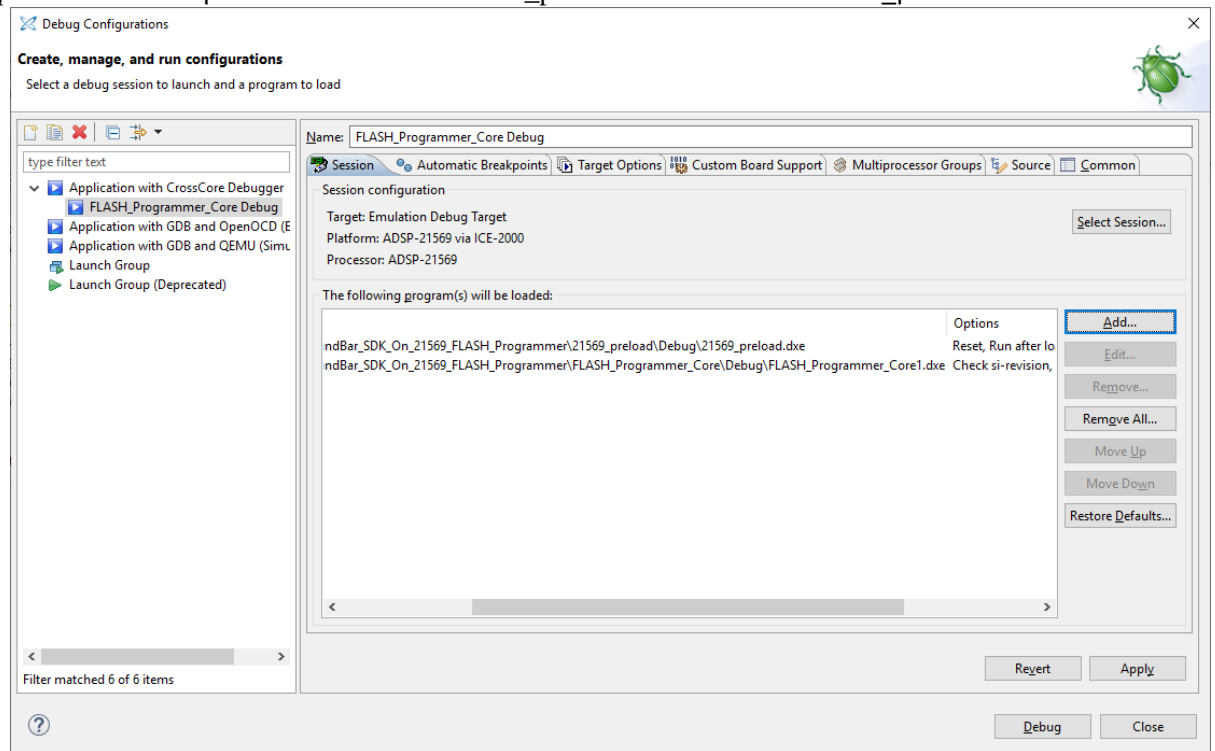


Step 3: Load the code using the “debug Configuration” option.



Step 4: Select the processor as “ADSP21569” and ICE1000/2000.

Step 5: Select the preload executable “21569_preload.dxe” from the 21569_preload folder

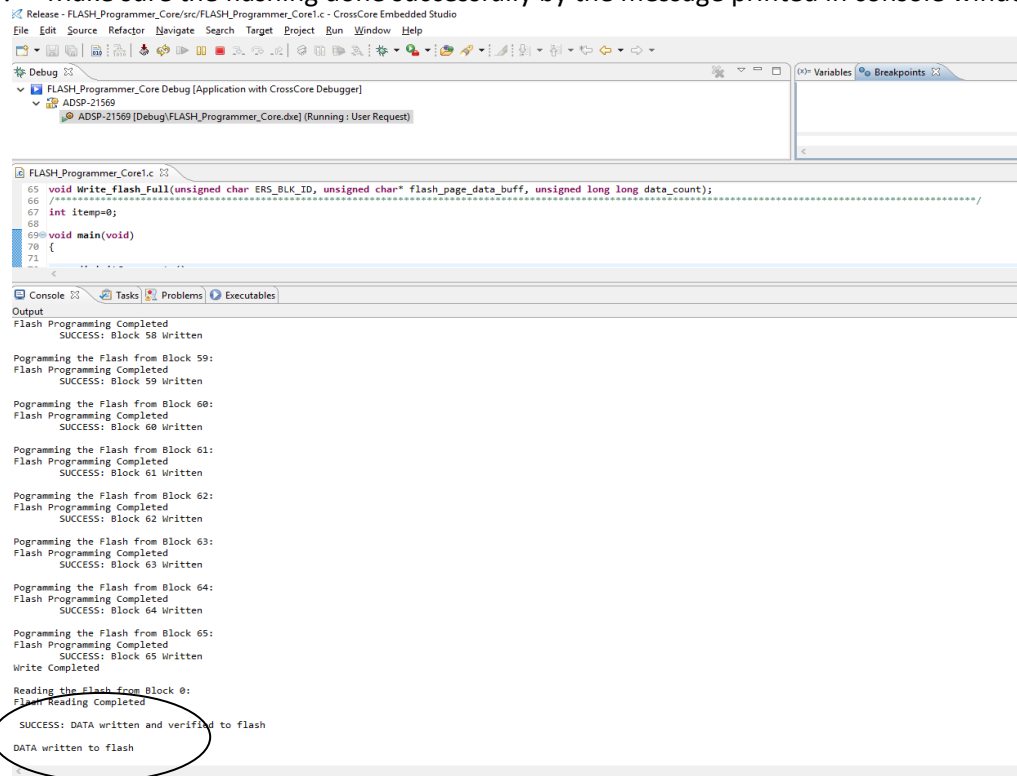


Step 6: Select Apply and Debug

Step 7: Select “Yes” for code loading message and debug message

Step 8: After the Load complete, “RESUME” the CCES and check the output window for the Flash writing progress

Step 9: Make sure the flashing done successfully by the message printed in console window.



Step 10: Connect the Jumper between 8 and 9 to J23 port as shown in below image.

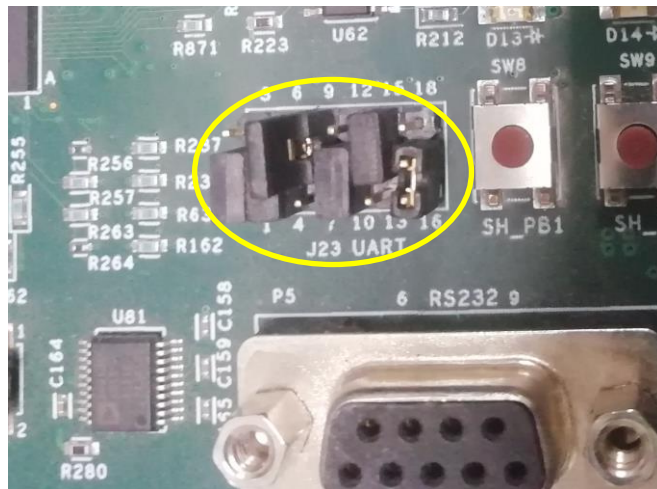


Figure 1: Jumper Setting